

Title-and-Tag Contrastive Vision-and-Language Transformer for Social Media Popularity Prediction

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ABSTRACT

Social media is an indispensable part of modern life, and social media popularity prediction (SMPP) plays a vital role in practice. In current work, the inconsistency of words in labels and titles, user feature transformation, etc have not been well noticed. In this paper, we propose a novel approach named Title-and-Tag Contrastive Vision-and-Language Transformer (TTC-VLT), combining two pre-trained vision and language transformers and other two dense feature parts for this prediction task. On one hand, in order to learn the differences between titles and tags, we design title-tag contrastive learning for title-visual and tag-visual, which separately extracts multimodal information from two types of text. On the other hand, user identification features are transformed to embedding vectors to capture user attribute details. From the extensive experiments, our approach outperforms the other methods on the social media prediction dataset. Our team achieve the 2nd place on the leader board of the Social Media Prediction Challenge 2022.

CCS CONCEPTS

• **Computing methodologies** → *Unsupervised learning*; • **Information systems** → *Multimedia information systems*.

KEYWORDS

Multimedia; Vision and Language Transformer; Contrastive Learning; Transformer

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1 INTRODUCTION

Massive posts containing abundant information are generated every day, not all of which are paid evenly attention on. To improve user experiences, service effectiveness, and commercial profits, reasonable analysis of posts is necessary. In the social media popularity prediction (SMPP) task of Social Media Prediction (SMP) Challenge 2022, a model based on the social media prediction dataset (SMPD) [21, 22, 24] is needed to accurately predict the popularity of each post collected from Flickr. The challenge is that the popularity of a post can be affected by various types of information in SMPD which are hard to be represented properly, including images, titles, tags, user profiles, etc.

In the last few years, various studies of SMPP have been proposed. On one hand, the works of [4, 8, 26] use deep neural networks (DNNs) to extract and fuse multimodal features. On the other hand, methods based on gradient boosting regression tree are also popular [1, 7, 9]. Furthermore, a feature generalization framework, proposed by [20], achieves feature generalization and temporal modeling to extract multi-modal features for SMPP.

Though the above mentioned methods show good performance in previous SMP Challenges, there are still 2 issues. First off, the difference between titles and tags is usually overlooked, which undoubtedly affect the accuracy of SMPP. Besides, training a DNN on the significantly large dataset, i.e., SMPD, with random initialization is essentially a difficult task which may cost vast time and computational resources to obtain a bad model.

In this paper, a novel approach named Title-and-Tag Contrastive Vision-and-Language Transformer (TTC-VLT) is proposed for SMPP. To tackle the problems caused by the difference between titles and tags, we develop a new unsupervised learning framework,

Title-and-Tag Contrastive learning. For more effective training, pre-trained model based on vision and language transformer (ViLT) is adopted to extract both image and text features. In terms of feature engineering, the identification embedding and statistical dense features mechanism are used for obtaining ample information from user attributes. Furthermore, statistical dense features and sparse features are processed separately.

The main contributions of this work are as follows:

- A concise single-stream architecture pre-trained model called ViLT is adopted, which performs well in SMPP tasks. To enhance the learning performance, two downstream tasks are used to fine-tune ViLT. In addition, pre-training and fine-tuning reduce quite a few time and computational costs.
- Contrastive learning is used between the titles and tags for higher accuracy. 2 two-stream ViLT models are built for title-visual and tag-visual, aiming to extract the multimodal information from two types of text.
- Effectiveness of the proposed method is evaluated by conducting comparison and ablation experiments on SMPD.

The remaining parts of this paper proceed as follows. In Section 2, previous research studies on SMPP are stated. Our approach is elaborated in Section 3. In Section 4, our approach is evaluated through simulations, compared with other approaches. Finally, we conclude our work in Section 5.

2 RELATED WORK

In recent years, various researches on how to predict popularity based on social media data have been undertaken. This paper focuses on leveraging Flickr social media data in the form of texts and images to predict the popularity of Flickr posts. We design our model by investigating various conventional multimodal methods and vision and language pre-training (VLP) methods.

2.1 Conventional Multimodal Methods for SMPP

Conventional multimodal learning methods are largely designed to help users handle specific vision-language features in SMPP tasks. The vision-language domain encompasses a broad range of tasks, including image captioning, visual question and answer, visual reasoning and image-text matching, etc. For example, Frome et al. train a deep visual-semantic embedding model to recognize visual objects, which uses both labeled image data and semantic information gained from text without annotation [5]. Xu et al. propose a model for generating image descriptions based on the attention mechanism [25]. Yang et al. propose stacked attention networks which use the semantic representation of a question as the query to search for the regions in an image that are related to the answer [27]. Malinowski et al. propose neural-image-QA, an end-to-end architecture combining a convolutional neural network and an long short-term memory (LSTM) for answering questions about images [15].

It is worth mentioning that, there are some effective methods proposed in the SMP challenge in 2019 and 2020. Ding et al. use pre-trained deep neural networks (DNNs), to extract the image and text features from diverse sources, which are then fed into a regressor [4]. Hsu et al. encode the image semantic features, which

are extracted by image captioning, and the text information using a trained word-to-vector model, which performs ensemble regression [8]. Xu et al. propose a multimodal deep learning framework that introduces the attention mechanism [26]. Furthermore, Wang et al. propose a hybrid framework for multimodal feature extraction, feature generalization and temporal modeling based on the sliding window mechanism [20]. Besides, Kang et al. propose a framework based on two Catboost regressors, one of which implements the feature representation and another serves as a regression model [9].

Furthermore, some winners use GBRT-based methods for prediction. Chen et al. propose a framework which contains visual-textual feature extraction, feature fusion and XGBoost regression for SMPP [1]. He et al. extracts textual features, numeric features, and user-related features, and then feeds the multiple features into LightGBM to predict popularity scores [7]. Kang et al. proposes a framework based on two Catboost regressors, one of which implements the feature representation and the other serves as a regression model [9].

2.2 VLP Methods for SMPP

As the paradigm of pre-training and fine-tuning has gained popularity, VLP shows the competitive performance on a variety of joint vision and language downstream tasks, including SMPP tasks. For instance, Lu et al. propose vision-and-language BERT (ViLBERT), a model for learning task-independent joint representations of images and natural languages [14]. ViLBERT builds on BERT as a multimodal dual-stream model, processing image and text inputs separately before interacting with them via the co-attention transformer layers. In addition, Li et al. introduce object-semantics aligned pre-training (Oscar), a multimodal pre-training method that uses objects as anchor points at the visual and linguistic-semantic levels to simplify the learning task of semantic alignment between images and texts [13]. Contrastive language-image pre-training (CLIP) [18] is proposed by Radford et al. as a method of learning from natural language supervision that learns to perform on over 30 distinct computer vision datasets. The experiments show that CLIP is more robust and computationally efficient than a supervised Image Net model of equivalent accuracy. Kim et al. propose vision-and-language transformer (ViLT), a minimal VLP model where the processing of visual inputs is simplified to a convolution-free manner in the same way as text processing inputs, to significantly facilitate pre-training [10].

3 METHODOLOGY

3.1 Model Overview

TTC-VLT is an end-to-end multimodal model composed of two pre-trained ViLT [10] models: sparse ID features part and statistical dense features part. The model structure is shown in Figure 1. The two pre-trained ViLT models contain title-visual transformer and tag-visual transformer. After pooling, the outputs of the two ViLT models are concatenated with embedding vectors of Sparse ID features and statistical dense features.

ViLT is a single-stream model which has a concise architecture as a VLP model with a minimal visual embedding pipeline. Image text matching (ITM) task and masked language modeling (MLM)

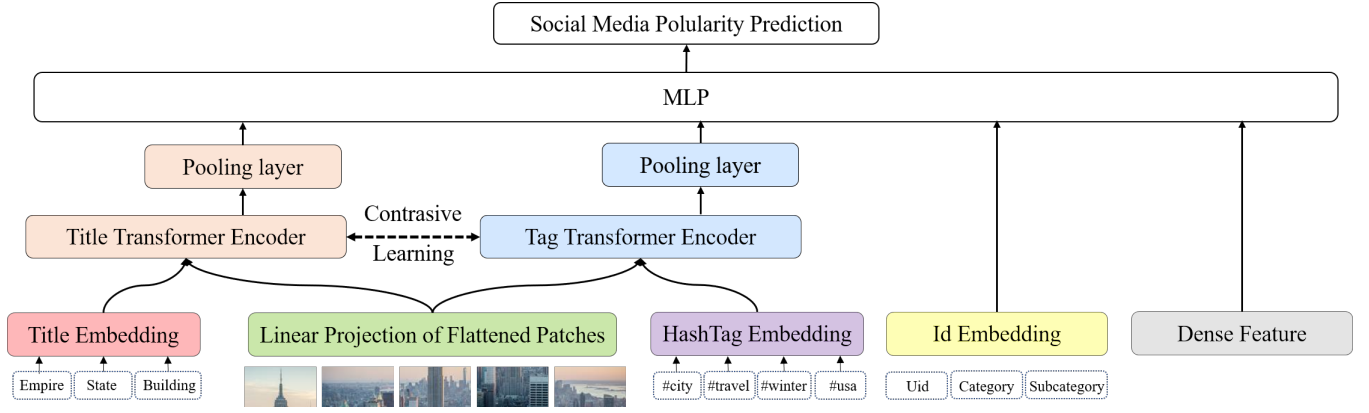


Figure 1: The structure of the Title-and-Tag Contrastive Vision-and-Language Transformer

task are used to pre-train ViLT. It applies image augmentations like RandAugment [3] except color inversion and cutout during fine-tuning. In TTC-VLT, no more pre-trained tasks are added for unsupervised learning. However, during fine-tuning, unsupervised contrastive learning and supervised learning are adopted for SMPP, in which the contrastive learning aims to constrain the gap between title-visual information and tag-visual information. The details will be provided in Section 3.2.

In SMPP, there are a lot of features like user characteristics, profiles, and term frequency-inverse document frequency (TF-IDF) features for titles and tags. We crawl additional user information from the user page indicated by “pathalias” [11] and build some user features like “FollowingCount”, “TotalPhoto”, etc.. Besides, we calculate some features about users’ images information such as the frequency that images of users have been reviewed. These dense features play an important role in our model.

Dense features are all numerical features, whose numerical values have certain meanings. However, features like user ID and category are sparse features. These features can not be directly put into MLP layers. These sparse features are transformed into embedding vectors for simple search and then inputted to MLP layers.

3.2 Title-Tag Contrastive Learning

Titles and tags are both text information, indicating that their feature spaces are consistent. Though text information has the capability of representing image information, there are still some differences between them. Firstly, titles in social media are ordered, which means when the order of words is changed, the titles will become hard to read or even unreadable. However, words in tags are separated, indicating the order of words is unfixed. The tags are readable even if their order is swapped. Secondly, there are many abbreviations in tags just like #MAGA (Make America Great Again) and #tbt (throw back thursday). These abbreviations fairly appear in the titles. Furthermore, their lengths are different. The average length of titles is 1.5 times longer than that of the tags. ViLT limit the length of the text to 20, which is sufficient for common visual-language tasks like image captioning and visually grounded

story generation. However, it is shorter in SMPP tasks when concatenating titles and tags together. Based on the above, two-stream training on titles and tags is seemingly a feasible decision.

We build 2 two-stream ViLT models for title-visual and tag-visual, which separately extract multimodal information from two types of text. There are some slight differences during the training phases. When one tag is inputted, the position embedding is the same. The input tag $t_{\text{tag}} \in \mathbb{R}^{L \times |V|}$ is embedded to $\tilde{t}_{\text{tag}} \in \mathbb{R}^{L \times H}$ with a word embedding matrix $T \in \mathbb{R}^{|V| \times H}$ and a constant position embedding $T^{\text{pos}=0} \in \mathbb{R}^{1 \times H}$. However, one title $t_{\text{title}} \in \mathbb{R}^{L \times |V|}$ is inputted, it should be embedded to $\tilde{t}_{\text{title}}$ with varied position embedding $T^{\text{pos}} \in \mathbb{R}^{(L+1) \times H}$, which means:

$$\tilde{t}_{\text{tag}} = [t_{\text{class}}; t_{\text{tag}_1} T; \dots; t_{\text{tag}_L} T] + T^{\text{pos}=0}, \quad (1)$$

$$\tilde{t}_{\text{title}} = [t_{\text{class}}; t_{\text{title}_1} T; \dots; t_{\text{title}_L} T] + T^{\text{pos}}. \quad (2)$$

The input image $I \in \mathbb{R}^{C \times H \times W}$ is sliced into patches and then flattened to $v \in \mathbb{R}^{N \times (P^2 \cdot C)}$, where (P, P) is the patch resolution and $N = HW/P^2$. Followed by linear projection $V \in \mathbb{R}^{(P^2 \cdot C) \times H}$ and image position embedding $V^{\text{pos}} \in \mathbb{R}^{(N+1) \times H}$, $\bar{v}$ is defined as:

$$\bar{v} = [v_{\text{class}}; v_1 V; \dots; v_N V] + V^{\text{pos}}. \quad (3)$$

The title, the tag, and image embedding vectors are summed with their corresponding modal-type embedding vectors t^{type} and v^{type} . Then, the summed vectors are concatenated into two combined sequences z_{tag} and z_{title} :

$$z_{\text{tag}} = [\tilde{t}_{\text{tag}} + t^{\text{type}}; \bar{v} + v^{\text{type}}], \quad (4)$$

$$z_{\text{title}} = [\tilde{t}_{\text{title}} + t^{\text{type}}; \bar{v} + v^{\text{type}}]. \quad (5)$$

The two combined sequences are separately passed into two stacked blocks that include multi-headed self-attention(MSA) layers, layer normalization(LN) and MLP layers, defined as:

$$\hat{z}_{\text{title}}^d = \text{MSA} \left(\text{LN} \left(z_{\text{title}}^{d-1} \right) \right) + z_{\text{title}}^{d-1}, \quad d = 1 \dots 12, \quad (6)$$

$$z_{\text{title}}^d = \text{MLP} \left(\text{LN} \left(\hat{z}_{\text{title}}^d \right) \right) + \hat{z}_{\text{title}}^d, \quad d = 1 \dots 12,$$

$$\hat{z}_{\text{tag}}^d = \text{MSA} \left(\text{LN} \left(z_{\text{tag}}^{d-1} \right) \right) + z_{\text{tag}}^{d-1}, \quad d = 1 \dots 12, \quad (7)$$

$$z_{\text{tag}}^d = \text{MLP} \left(\text{LN} \left(\hat{z}_{\text{tag}}^d \right) \right) + \hat{z}_{\text{tag}}^d, \quad d = 1 \dots 12.$$

However, the way of directly using the 2 streams does not consider the relationship between titles and tags. The 2 types of text can both describe images. To make a connection between title, tag, and images, we proposed a method named title-tag contrastive learning task in fine-tune processing. Title-tag contrastive learning task aims to estimate a lower bound of the mutual information between titles and tags. For each title x_{title} , we build a positive tag sample x_{tag}^+ which belongs to the same image. For each positive tag, it is associated with K random negative samples denoted as $[x_{\text{tag}_1}^-, x_{\text{tag}_2}^-, \dots, x_{\text{tag}_K}^-]$. To measure the non-normalized mutual information score between titles and tags, we apply a relevance function formulated as:

$$f(x_{\text{tag}}, x_{\text{title}}) = \frac{x_{\text{tag}}^T x_{\text{title}}}{\|x_{\text{tag}}\| \cdot \|x_{\text{title}}\|}. \quad (8)$$

Then, we apply InfoNCE [16] as our loss function, which formulated as follows:

$$\mathcal{L}_{CL} = -\log \left[\frac{e^{f(x_{\text{tag}}^+, x_{\text{title}})}}{e^{f(x_{\text{tag}}^+, x_{\text{title}})} + \sum_{i=1}^K e^{f(x_{\text{tag}_i}^-, x_{\text{title}})}} \right], \quad (9)$$

where $x_{\text{title}} = \tanh(z_{\text{title}}^{12} W_{\text{title}}^{\text{pool}})$ and $x_{\text{tag}} = \tanh(z_{\text{tag}}^{12} W_{\text{tag}}^{\text{pool}})$.

The input of the sparse features z_l is defined as:

$$x_l = e_{-m}^l(z_l), \quad (10)$$

where e_{-m}^l is the embedding matrix for the sparse features. The dense features x_d are directly inputted into the last MLP layer. We put all features into the last layer, which defined as:

$$\text{popularity} = \text{MLP}([x_{\text{title}}, x_{\text{tag}}, x_l, x_d]). \quad (11)$$

4 EXPERIMENT

4.1 Dataset

SMPD is a large-scale dataset for social media research. In SMPD, the contents of one post, including text information and image information, are greatly capable of reflecting popularity. Besides, attributes, such as user profile, temporal-spatial information, and category also have strong correlations with posts' popularity. It contains 3 levels of categories. The first level has 11 different classes. The second level contains 77 different classes. The third level contains 668 different classes.

4.2 Implementation Details

By quantitative evaluation, we adopt Spearman's Rho (SR) and the mean absolute error (MAE) to measure our model performance. In this subsection, we compare the performance of our method with baselines. The results are listed in Table 1.

Firstly, we try some methods like contextual LSTM (CLSTM)[6], visual-linguistic BERT (VL-BERT)[19], stacked cross attention network (SCAN)[12], and deep temporal context networks (DTCN)[23] as baselines. We also try the original ViLT[10]. ViLT, which is a single-stream model with a concise architecture as a VLP model with a minimal visual embedding pipeline. It initializes the interaction transformer weights from pre-trained ViT by exploiting the power of the interaction layers to process visual features. The inputs of ViLT, which consist of text embedding vectors and image

Table 1: The experiment on our method and report results. The SR and MAE of our model are compared with SCAN, VL-BERT, CLSTM, DTCN, CL-WMTG, CatBoost and ViLT.

Method	Hyperparameter	Offline	Online
SCAN[12]	–	0.53/1.84	–
VL-BERT[19]	head=1,layer=1	0.53/1.82	–
CLSTM[6]	–	0.54/1.80	–
DTCN[23]	–	0.63/1.52	0.62/1.50
ViLT[10]	head=12,layer=12	0.63/1.51	–/–
CL-WMTG[2]	head=1,layer=1	0.65/1.41	0.65/1.39
CatBoost[17]	–	0.68/1.40	0.69/1.35
TT-VLT	head=12,layer=12	0.70/1.40	0.70/1.42
TTC-VLT	head=12,layer=12	0.73/1.32	0.70/1.32
TTC-VLT+CatBoost	head=12,layer=12	0.75/1.22	0.71/1.27

embedding vectors summed with corresponding modal-type embedding vectors, are concatenated into a combined sequence. The model architecture of ViLT contains stacked blocks that include a 12-heads self-attention layer and a multilayer perceptron (MLP) layer whose size is 3072. In the above 4 methods, ViLT is the best. Compared with the other 3 methods, this method improves SR by at least 16% and reduces MAE by at least 15%.

Secondly, we make some improvements on the curriculum learning for wide multimedia-based transformer with graph target detection (CL-WMTG)[2]. We put features of wide part into CatBoost[17] and get a good online score. We obtain improvement of 6% over CL-WMTG.

At last, our TTC-VLT model gets the best performance among the previous mentioned models. To conduct some ablation study, we remove the contrastive learning method from TTC-VLT, and named it TT-VLT. Compared with ViLT, we add statistical dense feature and ID embedding features to the last hidden layer concatenated with title-visual transformer and tag-visual transformer. As a result, TT-VLT outperforms ViLT significantly, which shows that the sparse features and dense features greatly improve the performance of the model. However, the model performance is worse than TTC-VLT, which indicates that the unsupervised learning task does help. SR of TTC-VLT gets an improvement of 1.5%. We merge the two final results of TTC-VLT and catboost to get a better performance for online evaluation.

5 CONCLUSION AND FUTURE WORK

In this paper, we propose a novel approach named TTC-VLT for social media popularity prediction. Two pre-trained ViLT models are built to constrain the gap between titles and tags. Sparse ID features and statistical dense features are used to provide detailed information. We conduct extensive experiments in SMPD and win the 2nd place in SMP Challenge 2022. In the future, we will explore a better solution to capture more differences between features.

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